

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Third Semester B.E. (RI) (Credit System) Degree Examinations
 April 2023

21RI302 - INTRODUCTION TO ROBOTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

Marks BT* CO* PO*

- | | | | | | |
|----|--|---|-----|---|---|
| 1. | a) Explain any 3 types of Mechanical gripper mechanisms with simple sketches. | 6 | L*2 | 1 | 1 |
| | b) Define the resolution, accuracy and repeatability with respect to robot manipulator. | 6 | L1 | 1 | 1 |
| | c) Moving coordinate system is rotated about the X axis of the reference coordinate system through 180 degrees followed by translation along X, Y, and Z axis of the reference coordinate system by 5, 5 and 5 units respectively. Obtain the transformation matrix of moving coordinate system with respect to the reference coordinate system. If a point P [1,1,1] ^T is represented in the moving coordinate system, find the position of point P in the reference coordinate system. Also find the transformation matrix of reference coordinate system with respect to the moving coordinate system. | 8 | L3 | 1 | 1 |
| 2. | a) What are point to point robots and Continuous path robots? Explain. | 6 | L2 | 1 | 1 |
| | b) What is the selection and design consideration of grippers for robot applications? | 6 | L1 | 1 | 1 |
| | c) There is a reference coordinate frame {1} and a moving coordinate frame {2} which are coinciding initially. The coordinates of point P in frame {2} are [3, 2, 1] ^T . The moving coordinate frame {2} is now rotated about z-axis of the reference coordinate Frame{1} by 45 degrees. Find the coordinates of point 'P' with respect to the reference coordinate frame {1} after this rotation. | 8 | L3 | 1 | 3 |
| 3. | a) What is degree of freedom of a robot? What are the types of joints and joint notation scheme followed in robotics? Explain briefly. | 8 | L1 | 1 | 2 |
| | b) A robotic workstation has been setup with a TV camera as shown in Fig. Q3(b). The camera can see the robot coordinate system where base of a six joint robot is attached. It can also see the centre of an object (Cube) to be manipulated by the robot. If local coordinate system is established at the centre of the cube, this object as seen by the camera can be represented by a homogeneous transformation matrix ${}^{camera}T_{cube}$. If the origin of the base coordinate system as seen by the camera can also be expressed by the homogeneous transformation matrix ${}^{camera}T_{base}$. | | | | |

$${}^{camera}T_{cube} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 10 \\ 0 & 0 & -1 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{camera}T_{base} = \begin{bmatrix} 1 & 0 & 0 & -10 \\ 0 & -1 & 0 & 20 \\ 0 & 0 & -1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine the position and orientation of cube coordinate system with respect to the base coordinate system.

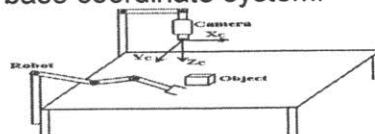


Fig. Q3(b)

P.T.O.

7 L3 1 3

- c) Frame {2} is rotated with respect to frame {1} about the X-Axis by an angle of 60 degree. The position of the origin of the frame {2} as seen from frame {1} is $[7 \ 5 \ 7]^T$. Obtain the transformation matrix 1T_2 which describes frame {2} relative to frame {1}. Find the transformation of 2T_1 . Also find the description of point P in frame {1} if ${}^2P = [2 \ 4 \ 6]^T$.

5 L2 1 3

Unit – II

4. a) Sketch and explain the hydraulic actuation system used in the robot. List merits and demerits of hydraulic actuation system.
b) What are velocity sensors? Sketch and explain following Velocity sensors used in the robotic applications.
i) Tachometer ii) Hall effect Sensor

10 L1 2 1

10 L1 2 1

5. a) With the help of a H-Circuit, sketch & explain the switching sequence for half stepping bipolar permanent magnet stepper motor.
b) Sketch and explain the working principle of Rotary absolute optical encoder. Also show the gray code and binary conversion.

10 L2 2 1

10 L2 2 1

6. a) What are Proximity sensors? Explain the working principle of following sensors:

i) Inductive proximity sensor

ii) Capacitive proximity sensor

8 L2 2 1

- b) Sketch and explain working principle and control of brushless DC Motor with switching circuit.

8 L2 2 1

- c) Define sensitivity and linearity with respect to a sensor.

4 L1 2 1

Unit – III

7. a) What are the advantages of using robots in industry? State any five industrial application of robot system.
b) Enumerate the differences between a Model Aircraft and a Drone.
c) What do you mean by an Unmanned Aerial vehicle and a Model Aircraft?

10 L1 2 1

6 L2 2 1

4 L1 2 1

8. a) What are the types of Medical Robots? Explain. What are the advantages and limitations of Medical Robots?
b) What is Legged Locomotion? Explain. What are its advantages over wheeled Locomotion?

10 L1 2 1

10 L1 2 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
